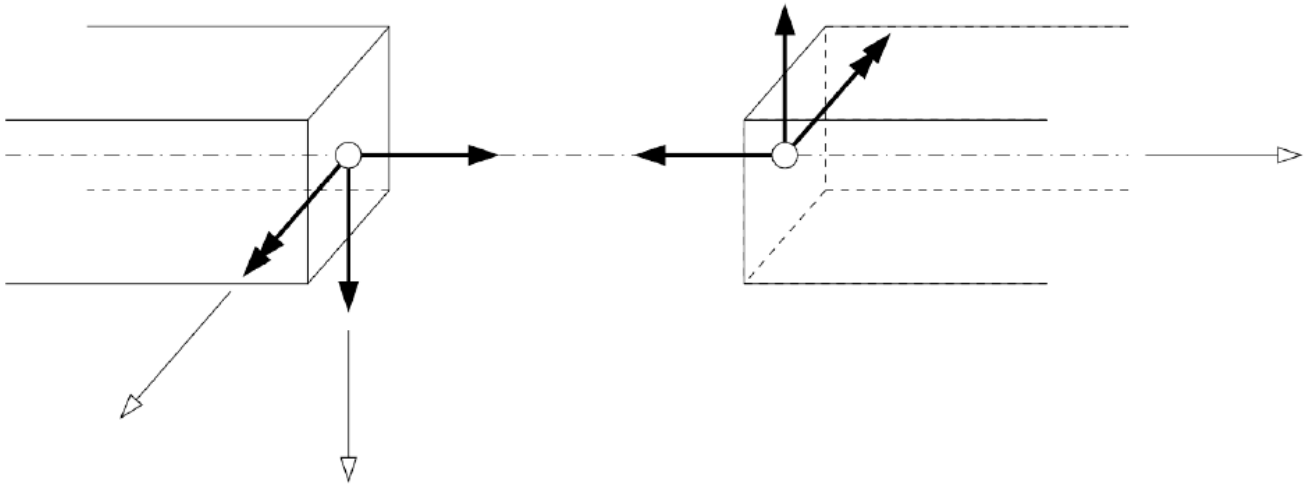
Engineering Mechanics I

Lecture

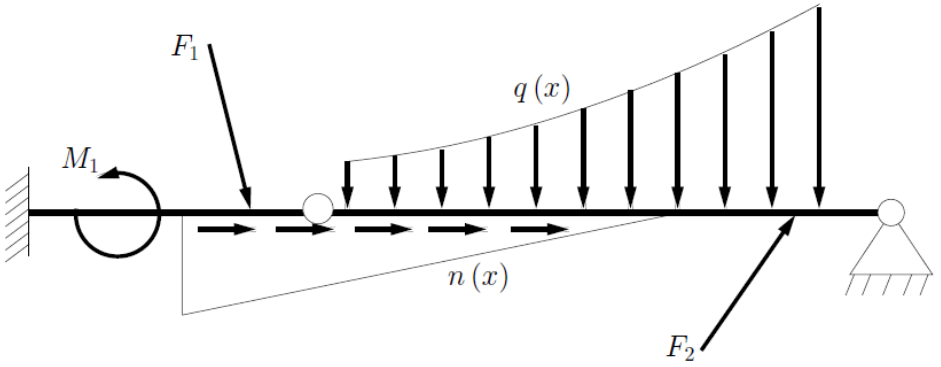


Autonomous Aerial Systems

7.5. Course of stress resultants along the beam

Step 2: Influence of individual forces and (individual) moments

- Do **not** "integrate over" places where structure or load change!
- System is to be broken down into sections.
- Corresponding transition conditions are to be formulated at section boundaries.



7.5. Course of stress resultants along the beam

Important Rules

- Without external forces
  - Shear force is constant
  - Bending moment is linear
- Without point loads
  - Slope of the moment line is Q, zero crossing of Q is condition for maximum of M
  - Curvature of the moment line is –q
- With point loads
  - Single force leads to jump in shear force line or normal force line and kink in moment line
  - Single moment leads to jump in moment line
- The superposition principle does always apply

7.5. Course of stress resultants along the beam

Graphical representation of internal forces

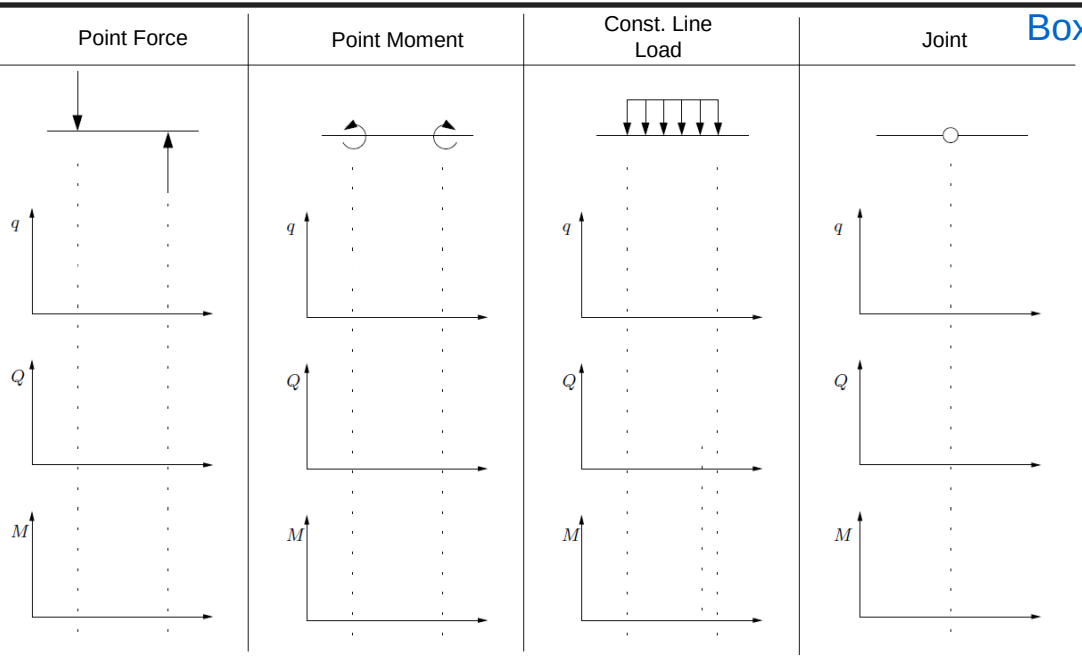
Essential for understanding load-bearing behavior!

Definition

- Trajectories with completely labeled coordinate system.
- Specify signs explicitly!
- Apply positive section size on the negative z-side (cf. literature), i.e. for horizontal elements mostly at the top.
- Distinguished points (section boundaries, extrema, nodes) are to be marked with function values as magnitude values.

7.5. Course of stress resultants along the beam

Box 7.19

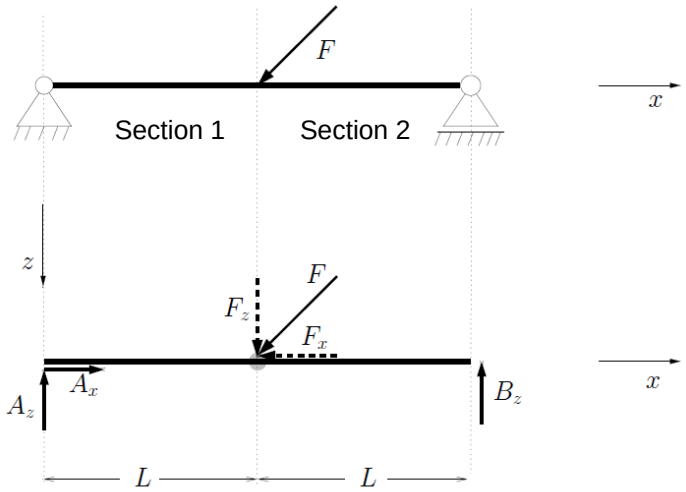


$q(x)$	< 0			$= 0$			> 0		
$Q(x)$	increasing			constant			decreasing		
	< 0	0	> 0	< 0	0	> 0	< 0	0	> 0
$M(x)$	left curved			straight			right curved		
	Decrea-sing	Min	Increa-sing	Decrea-sing	Horiz-ontal	Increa-sing	Decrea-sing	Max	Increa-sing

Autonomous Aerial Systems

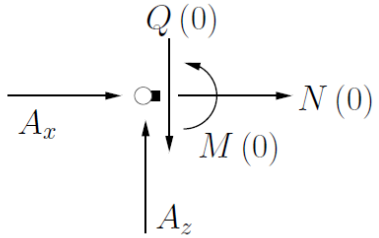
7.6. Examples

Example: Single span beam under point load



Box 7.20

7.6. Examples



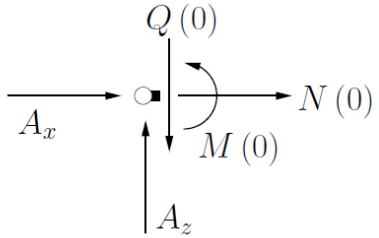
Section 1: x<L

Boundary conditions for internal forces: Support A

- Normal force:

Box 7.21

7.6. Examples



Section 1: x<L

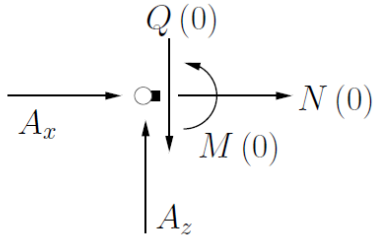
Boundary conditions for internal forces: Support A

- Normal force:

Box 7.21

- Shear force: $Q(x) = Q(0) - \int_0^x q(x)dx = \frac{1}{2}F_z = const.$

7.6. Examples



Section 1: x<L

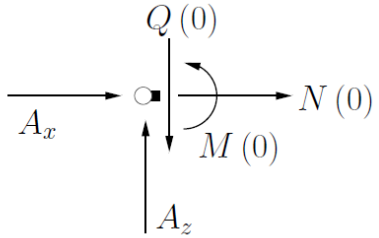
Boundary conditions for internal forces: Support A

- Normal force:

Box 7.21

- Shear force: $Q(x) = \underbrace{Q(0)}_{= A_z = \frac{1}{2}F_z} - \underbrace{\int_0^x q(x)dx}_{= 0} = \frac{1}{2}F_z = const.$

7.6. Examples



Section 1: x<L

Boundary conditions for internal forces: Support A

- Normal force:

Box 7.21

- Shear force: $Q(x) = \underbrace{Q(0)}_{= A_z = \frac{1}{2}F_z} - \underbrace{\int_0^x q(x)dx}_{= 0} = \frac{1}{2}F_z = const.$
- Moment: $M(x) = \underbrace{M(0)}_{= 0} + \int_0^x \underbrace{Q(x)}_{= \frac{1}{2}F_z} dx = \frac{1}{2}F_z \cdot x = linear$

Autonomous Aerial Systems

7.6. Examples

- **Section 2: L<x<2L**
- Boundary conditions for internal forces: Transition conditions

Box 7.22

- Normal force:

$$N(L+) \quad -\int_L^X n(x)dx = 0 = const.$$

- Shear force:

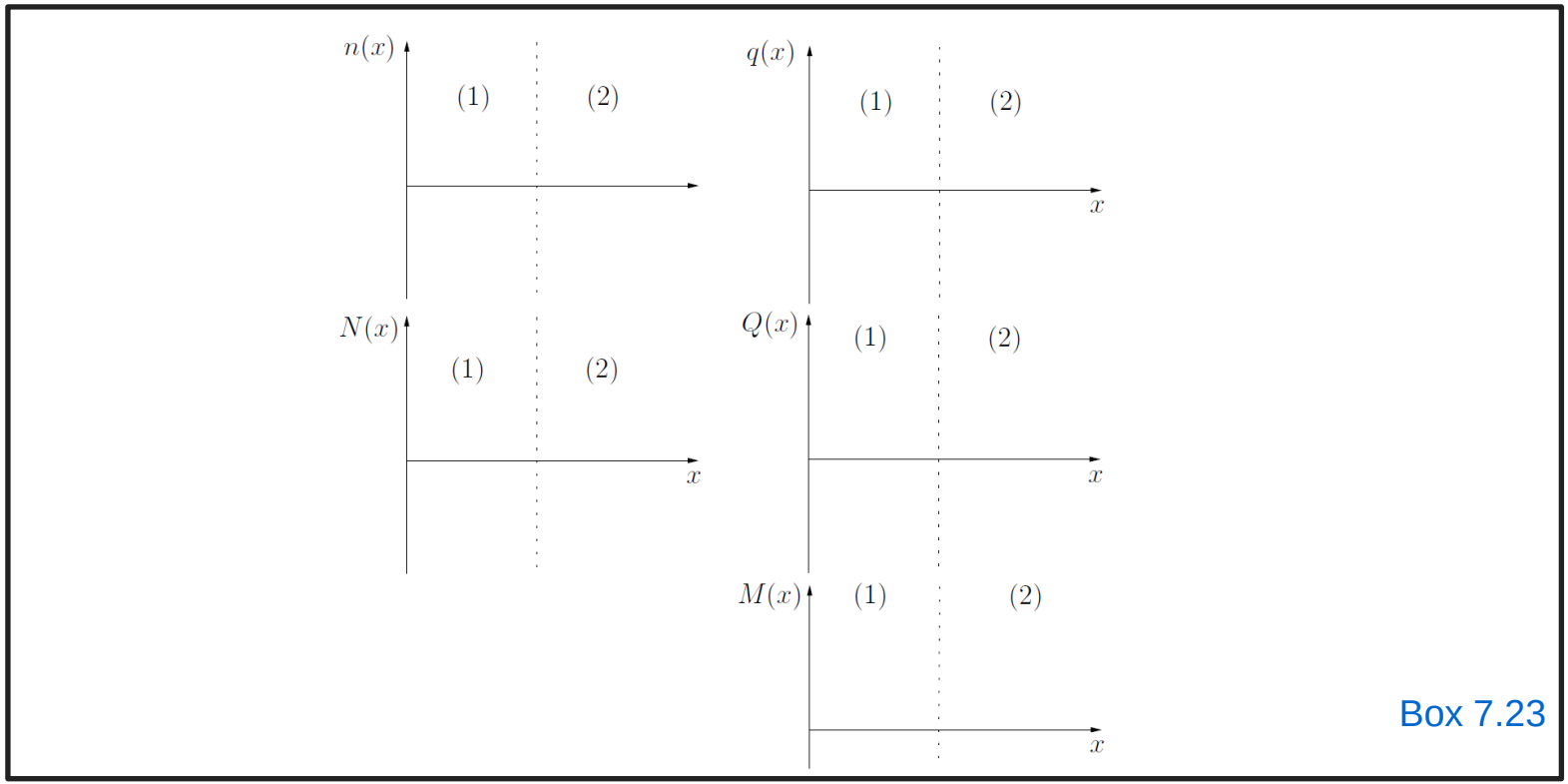
$$Q(L+) \quad -\int_L^X q(x)dx = -\frac{1}{2}F_z = const.$$

- Moment:

$$M(L+) \quad +\int_L^X Q(x)dx = -\frac{L}{2}F_z - \frac{1}{2}F_z(x-L) = \frac{1}{2}F_z(2L-x) = const.$$

Autonomous Aerial Systems

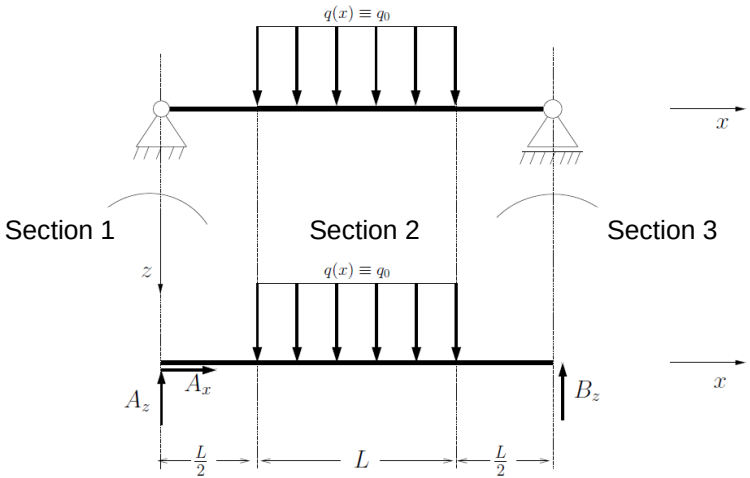
7.6. Examples



Box 7.23

7.6. Examples

Example: Single span beam with line load



Support forces:

$$\sum H = 0: \quad A_x = 0$$

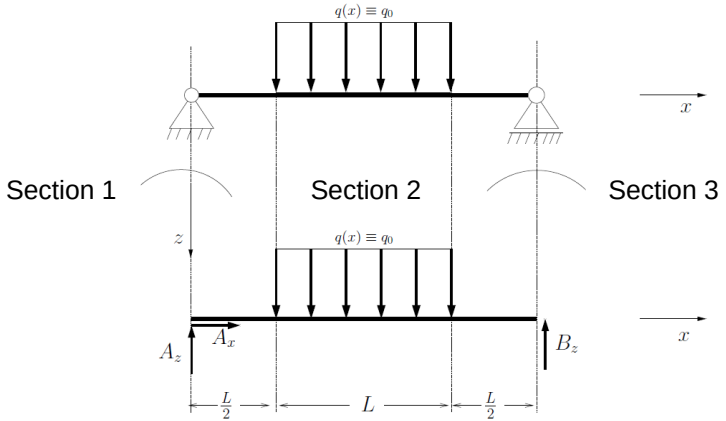
$$\sum M_A = 0: \quad B_Z \cdot 2L - q_0 L \cdot L = 0 \rightarrow B_Z = \frac{1}{2} q_0 L$$

$$\sum V = 0: \quad -A_Z - B_Z + q_0 L = 0 \rightarrow A_Z = \frac{1}{2} q_0 L$$

Check (Moment sum about B):

$$\sum M_B = 0: \quad A_Z \cdot 2L - q_0 L \cdot L = \frac{1}{2} q_0 L \cdot 2L - q_0 L \cdot L = 0$$

7.6. Examples



Integration over 3 sections:

- For Normal force:

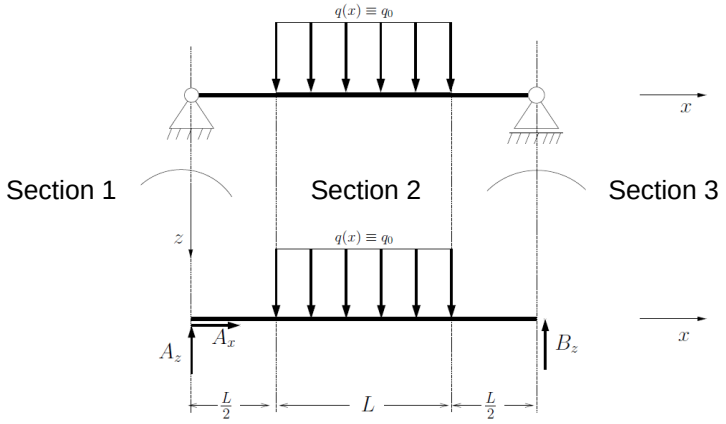
$$N(x) = 0 = const$$

.

- For Shear force (see Box below):

Box 7.24

7.6. Examples

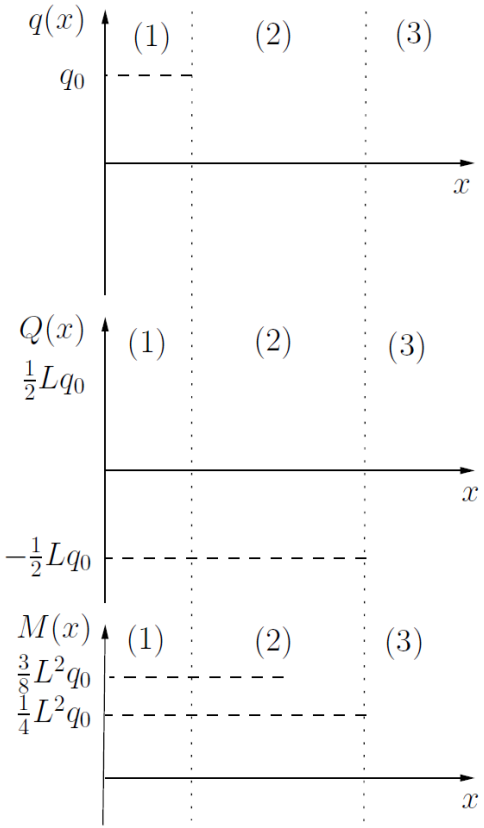


Integration over 3 sections:

- For the Moment

$$M(x) = \begin{cases} M(0) + \int_0^x \frac{1}{2} L q_0 dx = \frac{1}{2} L q_0 \cdot x & \text{for } x < \frac{1}{2}L \\ M\left(\frac{L}{2}+\right) + \int_{\frac{L}{2}}^x (L-x)\, q_0 dx = \frac{1}{4}L^2 q_0 - \\ \quad - \frac{1}{2} q_0 \cdot (L-x)^2 \Big|_{x=\frac{L}{2}}^x = \\ \quad = \frac{3}{8} L^2 q_0 - \frac{1}{2} q_0 \cdot (L-x)^2 & \text{for } \frac{1}{2}L < x < \frac{3}{2}L \\ M\left(\frac{3}{2}L+\right) + \int_{\frac{3}{2}L}^x \left(-\frac{1}{2} L q_0\right) dx = \\ \quad = \frac{1}{4} L^2 q_0 - \frac{1}{2} L q_0 \left(x - \frac{3}{2}L\right) & \text{for } \frac{3}{2}L < x < 2L \end{cases}$$

7.6. Examples



Box 7.25

Autonomous Aerial Systems